

# Internet of Things (IoT)



## IoT Network Access Technologies



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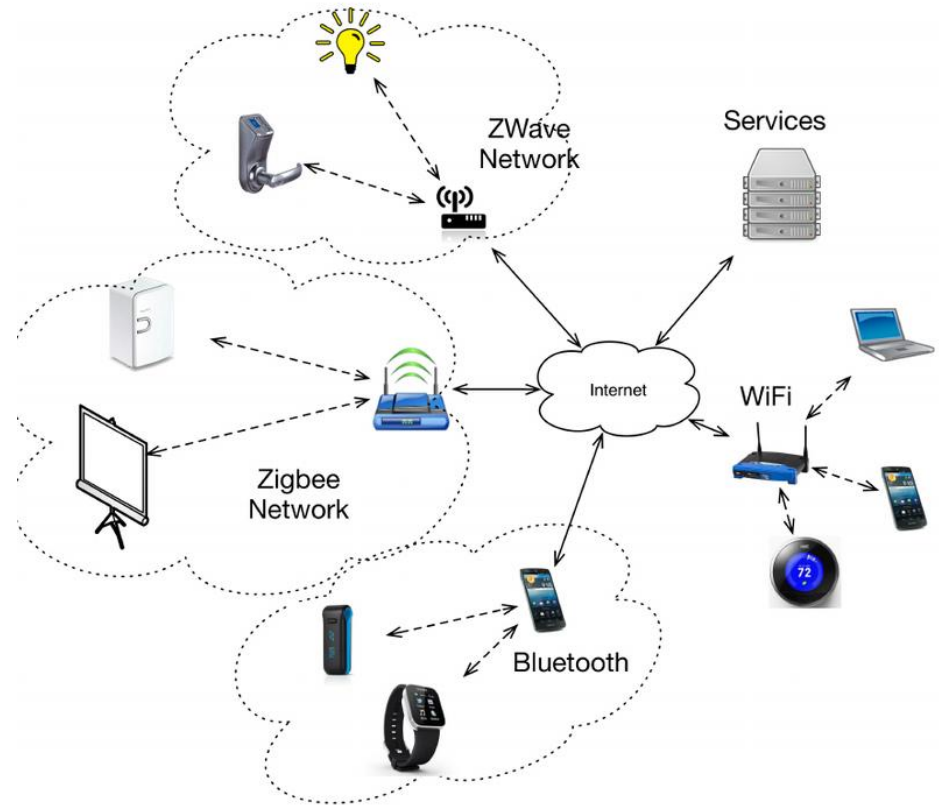
# IoT Network Access Technologies

- There are many IoT access technologies in the market today

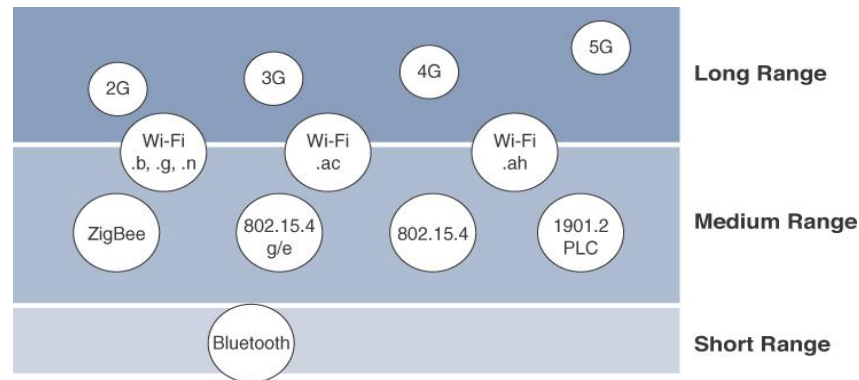


# Communications Criteria

- **Communication criteria** describes the **characteristics** and **attributes** of IoT access technologies
- IoT access technologies could be wired or wireless
- **Wireless communication is prevalent** for smart object connectivity
  - ease of **deployment**
  - allows smart objects to be **mobile**
  - moving without losing **connectivity**
- Few basic criteria:
  - Range
  - Frequency bands
  - Power consumptions
  - Topology
  - Constrained devices
  - Constrained-node networks



# Communication Range



## • Short range:

- **tens of meters** of maximum distance between two devices
- often considered as an alternative to serial cable
- Example:
  - IEEE 802.15.1 Bluetooth,
  - IEEE 802.15.7 Visible Light Communications (VLC)

## • Medium range

- **tens to hundreds of meters** between two devices
- **Wireless :**
  - IEEE 802.11 WiFi,
  - IEEE 802.15.4 Low Rate WPAN,
  - IEEE 802.15.4g Smart Utility Networks (SUN)
- **Wired :**
  - IEEE 802.3 Ethernet,
  - IEEE 1901.2 Narrowband Power Line Communications (PLC)

## • Long range

- **greater than 1 mile (1.6 km)** between two devices
- **Wireless :**
  - 2G, 3G, 4G,
  - IEEE 802.11ah,
  - Low-Power Wide-Area (LPWA) communications
- **Wired :**
  - IEEE 802.3 ethernet over optical fiber,
  - IEEE 1901.2 Broadband PLC

# Frequency Bands

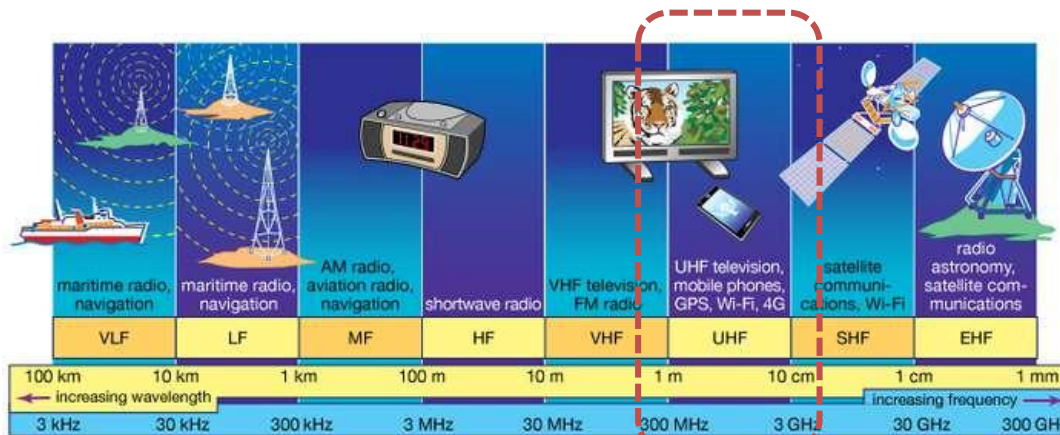
- Radio spectrum is **regulated by countries** and/or organizations
  - e.g. International Telecommunication Union (ITU), Federal Communications Commission (FCC), Telecom Regulatory Authority of India (TRAI)
- Frequency bands** leveraged by wireless communications are split between:

## 1. Licensed

- applicable to long-range access technologies
- users must subscribe to services
- common **licensed spectrum for IoT** :
  - Cellular (900-2100 MHz),
  - NB-IoT (700-900 MHz)

## 2. Unlicensed

- Unlicensed* means that no guarantees or interference protections are offered
- industrial, scientific, and medical (**ISM**) portions of the radio bands
- well-known **ISM bands for IoT** :
  - 2.4 GHz, 5 GHz, 915 MHz for WiFi, BLE, ZigBee;
  - 868 MHz for LoRa



## ISM Bands - Industrial, Scientific and Medical

**900MHz**  
vs.  
**2.4GHz**  
vs.  
**5GHz**

### 2.4GHz

#### **Advantages:**

- Higher bandwidth allows large data transfer, speed
- Components are smaller, cheaper

#### **Disadvantages:**

- Congested band due to abundance of Wi-Fi, Bluetooth, microwaves, cordless phones
- Attenuates much more quickly, will not pass through metal

### 900MHz

#### **Advantages:**

- More robust, less prone to interference
- Lower attenuation, travels further through more obstacles

#### **Disadvantages:**

- Low bandwidth prevents large data transfer, speed
- Components are larger at lower frequencies

### 5GHz

#### **Advantages:**

- Higher bandwidth allows large data transfer, speed
- Less congested, few RF devices in this band

#### **Disadvantages:**

- Low transmit power limitations
- High attenuation in cables, requires very high gain antennas

- **India** also allow **865-867 MHz** ISM band

# Power Consumption

## • Grid-powered node

- node has a **direct connection to a grid** power source
- **communications are not limited** by power consumption criteria
- ease of **deployment is limited** by the availability of a power source
- makes mobility more complex

## • Battery-powered nodes

- bring **more flexibility** to IoT devices
- batteries are **small in size**
- batteries **can be changed** or recharged

## • IoT Nodes

- Nodes are battery powered
- It's wireless access technologies must address
  - the needs of low power consumption
  - connectivity for battery-powered nodes



# Power Consumption

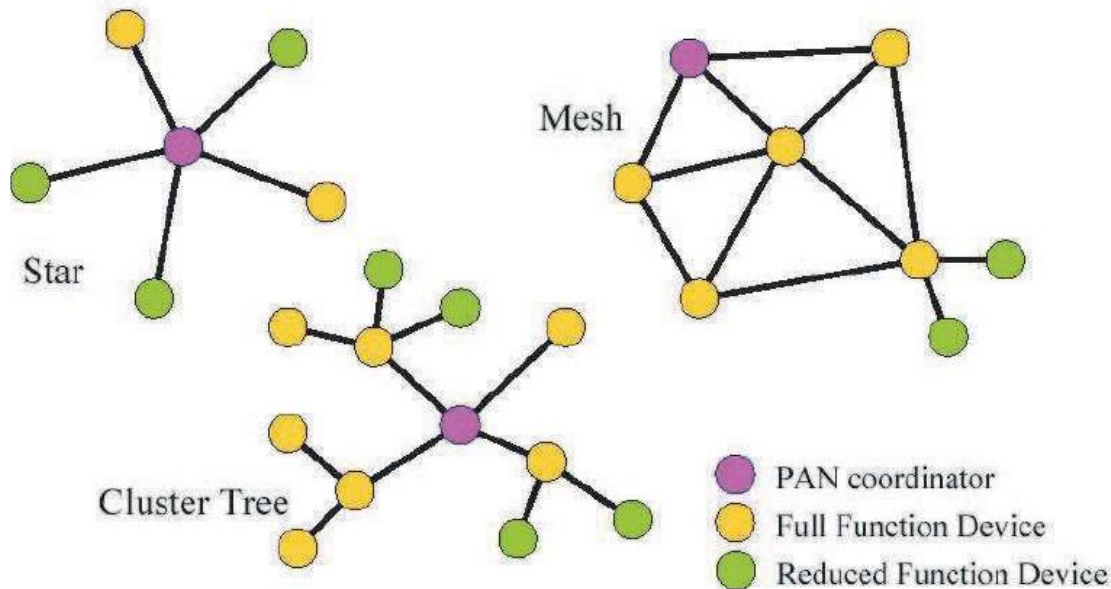
|              | Bluetooth     | ZigBee        | WiFi         | LoRaWAN     | NB-IoT      |
|--------------|---------------|---------------|--------------|-------------|-------------|
| Standard     | IEEE 802.15.1 | IEEE 802.15.4 | IEEE 802.11b | LoRaWAN     | 3GPP NB-IoT |
| Sleeping     | 9 $\mu$ A     | 12 $\mu$ A    | 30 $\mu$ A   | 0.1 $\mu$ A | 3 $\mu$ A   |
| Awake/Idle   | 35 mA         | 50 mA         | 245 mA       | 1.4 mA      | 6 mA        |
| Transmitting | 39 mA         | 52 mA         | 251 mA       | 44 mA       | 220 mA      |
| Receiving    | 37 mA         | 54 mA         | 248 mA       | 12 mA       | 46 mA       |
| Power Supply | 3.3 V         | 3.3 V         | 5 V*         | 3.3 V       | 3.6 V       |

\* The ESP8266 module powered by 3.3 V could be used as WiFi module.



# Topology

- Three main topology schemes are dominant:
    - star, mesh, and peer-to-peer (cluster tree)
  - For long-range and short-range technologies:
    - star topology is prevalent
  - For medium-range technologies:
    - star, peer-to-peer, or mesh topology is common
- IEEE 802.15.4, 802.15.4g, and wired IEEE 1901.2a PLC are generally deployed as a mesh topology.
  - Indoor Wi-Fi deployments are mostly star topologies



**FFD:** A node that implements the full network functions

**RFD:** The device can implement a subset of protocol functions to perform just a specialized part (communication with the coordinator).

# Constrained Devices

- Constrained nodes have limited resources that impact their networking feature set and capabilities.
- RFC 7228 defines three classes for constrained nodes: Class 0, 1, 2

|         | RAM     | Flash Storage | IP stack           | Security Scheme | Example     |
|---------|---------|---------------|--------------------|-----------------|-------------|
| Class 0 | < 10 KB | < 100 KB      | Not present        | No              | Push button |
| Class 1 | > 10 KB | > 100 KB      | Optimized IP stack | Light           | Sensors     |
| Class 2 | > 50 KB | > 250 KB      | Full IP stack      | Yes             | Smart meter |

# Constrained Networks



- Constrained-node networks are often referred to as low-power and lossy networks (LLNs)
- **Layer 1** and **Layer 2 protocols** must be evaluated using the following characteristics:
  - data rate and throughput
  - latency and determinism
  - overhead and payload.
- **Data rate & throughput:**
  - data rates available from 100 bps to tens of Mbps
  - actual throughput is less, sometimes much less, than the data rate
- **Latency & determinism:**
  - When latency is a strong concern, emergent access technologies such as Time-Slotted Channel Hopping (**TSCH**) mode of IEEE 802.15.4e should be considered.
- **Overhead & Payload**
  - The minimum IPv6 MTU size is expected to be **1280 bytes**.
  - MTU size for IEEE 802.15.4 is **127 bytes**; payload in LoRaWAN may be from **19 to 250 bytes**
  - So, the **fragmentation** of the IPv6 payload has to be performed by the link layer

# Comparison of Access Technologies



|                                  | WiFi                   | BLE                                                  | Thread        | Sub-GHz: TI | SigFox  | ZigBee        | LoRa                      |
|----------------------------------|------------------------|------------------------------------------------------|---------------|-------------|---------|---------------|---------------------------|
| <b>Max. Data throughput</b>      | 72 Mbps                | 2 Mbps                                               | 250 Kbps      | 200 Kbps    | 100 bps | 250 Kbps      | 50 Kbps                   |
| <b>Range</b>                     | 100 m                  | 750 m                                                | 100 m         | 4 km        | 25 km   | 130 m         | 10 km                     |
| <b>Topology</b>                  | Star                   | P2P/<br>Mesh                                         | Mesh/<br>Star | Star        | Star    | Mesh/<br>Star | Star of Star              |
| <b>Frequency</b>                 | 2.4 GHz                | 2.4 GHz                                              | 2.4 GHz       | Sub-GHz     | Sub-GHz | 2.4 GHz       | Sub-1GHz                  |
| <b>Power consumption</b>         | 1 Year<br>(AA battery) | Up to years on a coin-cell battery for limited range |               |             |         |               | Few Years<br>(AA battery) |
| <b>IP at the device node</b>     | Yes                    | No                                                   | Yes           | No          | No      | No            | No                        |
| <b>Deployed Consumer Devices</b> | AP                     | smart phones                                         | No            | No          | No      | No            | No                        |

# Lessons Learned



- Different Attributes of Access Technologies in IoT
  - ✓ Communication Criteria
  - ✓ Communication Range
  - ✓ Frequency Band
  - ✓ Power consumption
  - ✓ Topology
- Classifications of Constrained Nodes
- Characteristics of Constrained Networks

# Thanks!



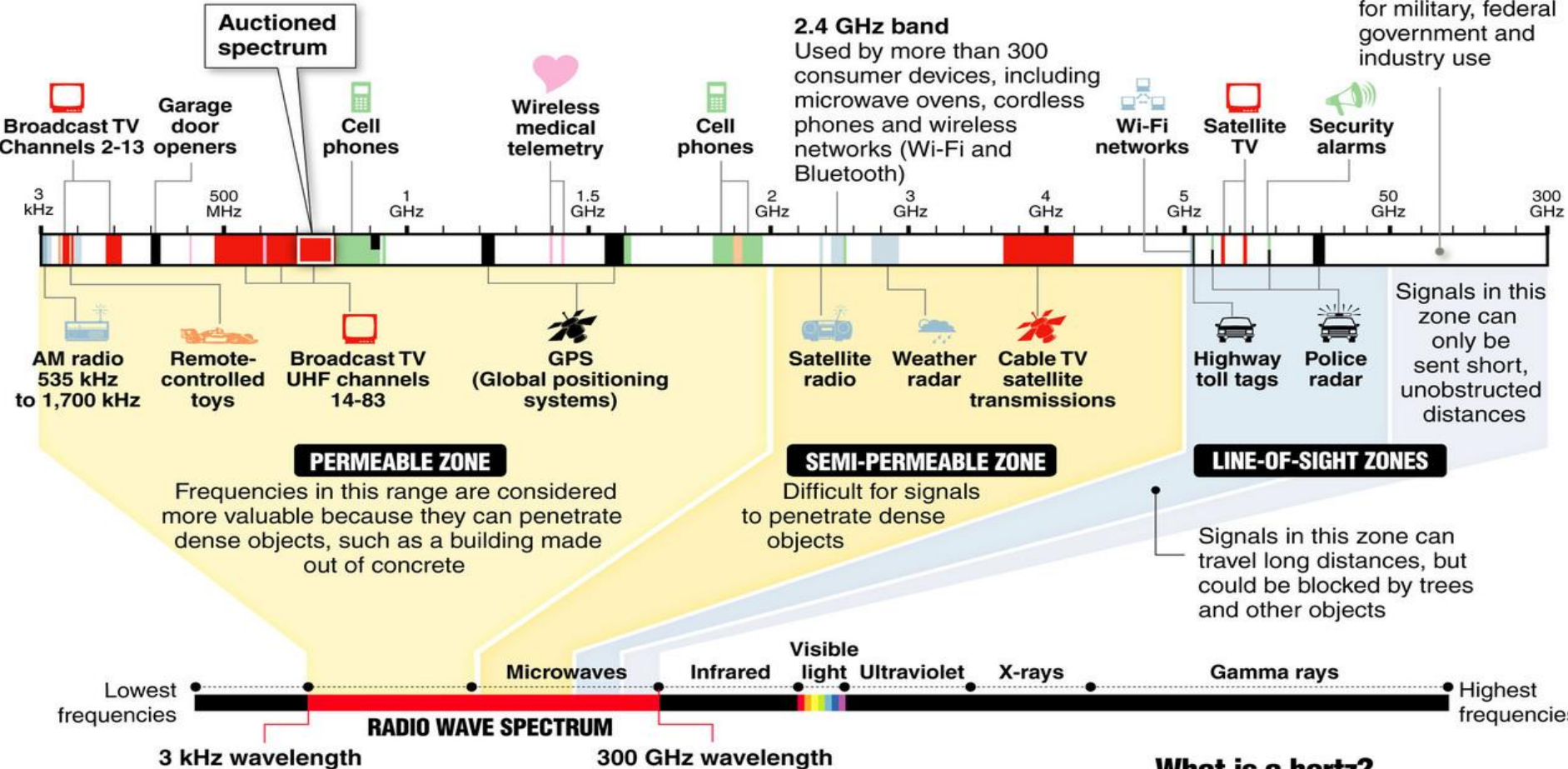
Figures and slide materials are taken from the following sources:

1. David Hanes *et al.*, “IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things”, 1<sup>st</sup> Edition, 2018, Pearson India.

# Inside the radio wave spectrum

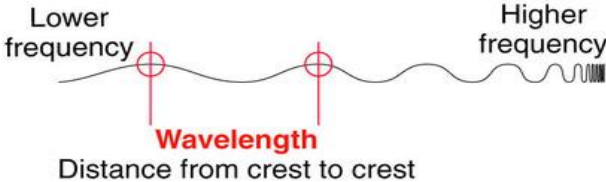
Almost every wireless technology – from cell phones to garage door openers – uses radio waves to communicate. Some services, such as TV and radio broadcasts, have exclusive use of their frequency within a geographic area. But many devices share frequencies, which can cause interference. Examples of radio waves used by everyday devices:

Most of the white areas on this chart are reserved for military, federal government and industry use



## The electromagnetic spectrum

Radio waves occupy part of the electromagnetic spectrum, a range of electric and magnetic waves of different lengths that travel at the speed of light; other parts of the spectrum include visible light and x-rays; the shortest wavelengths have the highest frequency, measured in hertz



## What is a hertz?

One hertz is one cycle per second. For radio waves, a cycle is the distance from wave crest to crest

- 1 kilohertz (kHz) = 1,000 hertz
- 1 megahertz (MHz) = 1 million hertz
- 1 gigahertz (GHz) = 1 billion hertz